Natalia Rabinovich

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Education

Rutgers University Honors College — New Brunswick, NJ
Bachelor of Science in Aerospace Engineering, Dean's List — *May 2028*

- **Relevant Coursework:** Fluid Mechanics, Thermodynamics, Mechatronics, CAD, Calc I – V
- **Technical Training:** UNP Satellite Fabrication Course

Relevant Experience

Program Manager — Sep 2024 – Present
SPICEsat CubeSat, Rutgers — *New Brunswick, NJ*

- Lead an 8-subteam, 40-member CubeSat program, coordinating mechanical, thermal, electrical, and software teams through spacecraft design, hardware integration, testing, and major indsutry-level technical reviews.
- Own Phase A execution within UNP's spacecraft development cycle, tracking engineering requirements, action items, schedules, technical documentation, and hardware risks through spacecraft design reviews.
- Lead weekly multidisciplinary design reviews to identify subsystem interface conflicts, assign action owners, and resolve technical risks before hardware testing, submission deadlines, and major program reviews.

Lead Systems Engineer (Incoming) — Fall 2026
NASA L'SPACE Mission Concept Academy — *Online*

- Selected as Lead Systems Engineer for NASA L'SPACE Mission Concept Academy, leading requirements development, subsystem interface definition, trade studies, and system-level technical coordination.

Undergraduate Research Assistant — Fall 2026
Space Propulsion with Advanced Chemistry and Energetics (SPACE) Lab, Rutgers — *New Brunswick, NJ*

- Conducting research in air breathing electric propulsion, working with VLEO wind tunnel, conducting simulations.

Thermal Engineering Project Lead — Summer 2026
SPICEsat CubeSat, Rutgers — *New Brunswick, NJ*

- Led a 4-member team developing the thermal-control strategy for a CubeSat fluid payload, translating 9 mission requirements into thermal limits, orbital analysis cases, and hardware design constraints for CDR readiness.
- Simplified spacecraft CAD in SolidWorks for thermal modeling by removing noncritical hardware and small geometric features while preserving key geometry, interfaces, contact surfaces, and component dimensions.
- Ran 10+ hot/cold case orbital simulations in Thermal Desktop and use MATLAB to process temperature histories, predicting payload temperatures from -10C to 40C, and improving fluid temperature distribution gradient by 20%.

Mechanical Lead — Sep 2022 – May 2024
Questionable Engineering Team, FIRST Robotics Competition — *Jersey City, NJ*

- Led mechanical design in Onshape/AutoCAD, achieving reliable arm motion and object manipulation in matches.

Projects

Autonomous Docking Robot — Spring 2026
Mechatronics Project

- Designed, built, and tested an autonomous docking robot using SolidWorks, 3D-printed hardware, an HC-SR04 sensor, DC motors, LEDs, and a servo-actuated clasp, using sensor to control speed and trigger docking sequence.

Mechanical Claw Machine — Spring 2026
SolidWorks CAD Project

- Designed a complete claw-machine assembly in SolidWorks, including the casing, gantry, claw, access door, shafts, guides, gears, and rack-and-pinion components needed for horizontal travel, vertical extension, and claw actuation.

Skills & Awards

Mechanical Design: SolidWorks, Onshape, AutoCAD, Fusion, FEA, CAD, 3D printing, mechanical prototyping
Hardware: Hand tools, soldering, wiring, staking, potting, conformal coating, cleanroom practices, hardware integration
Engineering: Trade studies, mechanical integration, technical documentation, PDR/CDR preparation, test planning
Programming: MATLAB, Python, C++, HTML, CSS, JavaScript
Software: Thermal Desktop, Microsoft Excel, PowerPoint, Word, Git, Linux
Languages: Fluent in English, Spanish, and Russian